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PROJECT MIRmade

Model Irregular Rhythms - Made with ♥

Project Brief

AI-Predictive Psychological Distress Modeling in Cardiac Arrhythmia Populations

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Project Definition

Project MIRmade (Model Irregular Rhythms - Made with ❤) is an open-source research initiative building AI/ML predictive systems that detect psychological distress in cardiac arrhythmia patients before crisis - closing the gap between the 15-40% who screen positive and the 2.5% who receive treatment.

Abstract

Cardiac rhythm disorders affect more than 100 million people worldwide. Among them, psychological distress - anxiety, depression, panic, and post-traumatic stress - is not a secondary concern. It is a co-occurring epidemic. Research consistently identifies positive screening rates of 15-40% for clinically significant anxiety and depression in arrhythmia populations, yet routine clinical screening captures only 2.5% of cases. The detection-to-treatment pipeline is broken at every level: screening is inconsistent, referral pathways are absent, and psychological expertise within electrophysiology clinics is virtually nonexistent.

In 2025, the European Society of Cardiology published its first-ever Clinical Consensus Statement on Mental Health and Cardiovascular Disease, formally recommending systematic mental health screening during cardiovascular care and calling for the creation of Psycho-Cardio Teams. The American Heart Association has issued parallel screening recommendations. The institutional mandate now exists. The technological infrastructure to answer it does not.

Project MIRmade aims to commit more compute to understanding how predictive modeling with modern AI/ML tooling - ECG foundation models, clinical natural language processing, wearable biosignal analysis, and digital phenotyping - can provide deeper understanding of psychological distress markers. This work is conducted with de-identified data, performed passively and privately, designed to avoid cue exhaustion and alert fatigue. The research identifies and observes data patterns that can align distress cues with an evolving digital cognitive behavioral therapy landscape, where interventions like those demonstrated by Särnholm et al. have produced 56% reductions in healthcare consumption when cardiac-related fear and avoidance are targeted.

Remarkable progress is being made in cardiac AI and in digital mental health independently. What remains underexplored is the intersection - applying the tools of one domain to the patient population of the other. MIRmade aims to contribute to this emerging area of research by assembling proven components from adjacent fields and applying them to the cardiac arrhythmia-psychological distress intersection, supporting and extending the work already underway in psychocardiology.

The Crisis in Numbers

Every statistic below is sourced from peer-reviewed, primary-source clinical research. These are the numbers OneRhythm exists to make visible - and the numbers Project MIRmade exists to move.

Metric	Rate	Source
Anxiety prevalence in arrhythmia patients	**88.3%**	Schnabel et al., Nature Sci Rep (2025)
Depression prevalence in arrhythmia patients	**71.1%**	JACC 2020 meta-analysis; Nature Sci Rep (2025)
Suicidal ideation in symptomatic AF patients	**20%**	Westermair et al., JAHA 2024; Kalman et al.
ICD shock → clinical anxiety odds ratio	**3.92×**	Sears et al.; EP Europace 2023 (N=39,954)
PTSD-related 5-year mortality risk (ICD)	**3.2×**	Edmondson et al., JACC 2013
Global AF/AFL patient population	**52.55M**	GBD 2021
Total global arrhythmia population (all types)	**>100M**	GBD + PSVT + channelopathy prevalence
Lifetime arrhythmia risk	**1 in 3**	EHRA 2025; Linz et al., Lancet Regional Health
Screening-positive vs. treated	**15–40% vs. 2.5%**	2026 PMC review; AHA screening data
Psych distress in “benign” arrhythmias (PVCs)	**40–60%**	Mikhaylov et al., Arch Med Sci (2020)

Note: This table is intentionally inclusive of all arrhythmia types. While AF represents the largest population volume, psychological distress is documented across the full spectrum - from SVTs and WPW to ventricular arrhythmias, PVCs, and inherited channelopathies. No rhythm is excluded.

No Rhythm Is Excluded

The psychological burden of arrhythmia is not limited to atrial fibrillation. Patients with supraventricular tachycardias (SVT) live with unpredictable racing heart episodes that mimic panic attacks - and are frequently misdiagnosed as panic disorder, compounding the psychological harm with medical invalidation. Patients with premature ventricular contractions (PVCs), often labeled “benign” by their physicians, report anxiety and depression rates of 40-60%. Patients with implantable cardioverter-defibrillators (ICDs) carry a device that can deliver painful shocks without warning - a 3.92-fold increase in clinical anxiety risk with each shock event. Patients with inherited channelopathies like Long QT syndrome and Brugada syndrome navigate the knowledge that their heart rhythm could become fatal at any moment. Patients recovering from catheter ablation face a psychologically vulnerable blanking period where recurrence anxiety peaks.

If you live with any kind of irregular heartbeat - whether it races, skips, flutters, pauses, or shocks - the psychological weight of that experience is real, it is measurable, and it is being largely ignored by the healthcare systems designed to help you. This project exists because that is unacceptable.

The Economic Case: GDP Impact and Healthcare Costs

The urgency of addressing psychological distress in arrhythmia patients is not only clinical - it is economic. The financial burden of untreated cardiac-psychological comorbidity falls on patients, employers, insurers, and entire national economies.

Direct Healthcare Costs

A 2026 study published in *JAMA Network Open* (Zhang et al.) analyzed privately insured U.S. employees aged 18-64 and found that each patient with atrial fibrillation incurs approximately **$11,393 in excess annual medical costs** compared to matched controls - driven by emergency department visits, inpatient admissions, outpatient visits, and prescription fills. Total U.S. treatment costs for atrial fibrillation alone exceed **$6.65 billion per year**, with approximately 75% attributable to hospitalization. In the UK, AF care already accounts for roughly 1% of the entire NHS budget - exceeding the combined cost of several common cancers.

When psychological distress goes undetected and untreated in these patients, costs compound. Depressed cardiac patients use more emergency services, require more repeat procedures, and have significantly higher 30-day readmission rates. The CMS Hospital Readmissions Reduction Program penalizes hospitals up to 3% of base DRG payments for excess readmissions - a direct financial consequence of the treatment gap MIRmade is designed to help close.

Productivity Losses and GDP Impact

The same JAMA Network Open study documented approximately **$840 in annual lost productivity per working AF patient** attributable to sick leave, short-term disability, and long-term disability. With an estimated 6-12 million AF patients in the United States alone (and over 52.55 million globally), the aggregate productivity loss is staggering. A 2025 Global Burden of Disease analysis focused specifically on AF/AFL in working-age populations (ages 30-64) found that age-standardized incidence and prevalence rates have been steadily increasing since 1990, directly amplifying economic productivity losses.

These figures capture only the cardiac burden. The psychological overlay - when a patient with arrhythmia also develops anxiety, depression, or PTSD - drives additional absenteeism, presenteeism (reduced effectiveness while at work), and early workforce exit. Employees with arrhythmias have been shown to have significantly higher costs, absences, and objective productivity losses compared to matched controls.

The Reactivation Argument

Beyond cost avoidance, there is a productivity recovery argument. When psychological distress in arrhythmia patients is identified early and treated effectively - as demonstrated by Särnholm et al.'s 56% reduction in healthcare consumption through targeted CBT for cardiac-related fear and avoidance - patients return to fuller participation in work and life. The Social Security Administration recognizes recurrent arrhythmias as a qualifying disability under Section 4.05 of its Blue Book. For patients whose disability is driven or amplified by untreated psychological distress rather than the arrhythmia itself,

effective mental health intervention could mean the difference between workforce participation and workforce exit. At a population scale, this represents a meaningful GDP recovery opportunity - reactivating segments of the labor force currently sidelined not by their hearts, but by the fear, anxiety, and depression that surround their hearts.

The Treatment Gap

The 15-40% vs. 2.5% treatment gap is not primarily a screening failure. It is an implementation and delivery failure - and that is precisely the kind of gap AI systems are built to close.

Research studies using validated instruments (PHQ-9, GAD-7, Cardiac Anxiety Questionnaire) consistently identify positive screening rates of 15-40% for anxiety and depression in arrhythmia populations. Routine EHR-based clinical screening captures only 2.5% positive screens. The detection-to-treatment pipeline is broken at every level: screening is inconsistent, referral pathways are absent, and psychological expertise within electrophysiology clinics is virtually nonexistent. A 2025 review in *Current Cardiology Reports* concluded that systematic screening and the use of patient-reported outcome measures remain inconsistent, leaving many psychological needs unaddressed.

Institutional Validation: The ESC and AHA Consensus

The 2025 European Society of Cardiology Clinical Consensus Statement on Mental Health and Cardiovascular Disease represents a watershed moment for psychocardiology. For the first time, the ESC formally recommends systematic mental health screening during cardiovascular care, introduces the ACTIVE framework (Acknowledge, Check, Tools, Implement, Venture, Evaluate), and calls for the creation of multidisciplinary Psycho-Cardio Teams. The American Heart Association has issued parallel screening recommendations.

This consensus creates institutional demand for new tools and approaches. Project MIRmade aims to contribute research and open-source infrastructure in support of the ESC's vision for integrated psycho-cardio care.

MIRmade's Approach: More Compute, Better Outcomes

MIRmade treats the treatment gap as an engineering problem amenable to modern AI/ML approaches. The project assembles proven technologies from adjacent domains - ECG foundation models pretrained on over 10 million recordings, clinical NLP models trained on millions of clinical notes, wearable biosignal processing validated in panic attack prediction - and applies them to the cardiac arrhythmia–psychological distress intersection, building on the foundational clinical research of groups like Särnholm, Lampert, and the ESC psychocardiology working groups.

The approach is designed around three critical constraints. First, all analysis uses **de-identified data**, processed passively and privately, respecting the heightened protections required for mental health information under HIPAA, 42 CFR Part 2, and GDPR Article 9. Second, the system is designed to **avoid cue exhaustion** - the three-tier alert architecture ensures that only high-confidence, high-severity predictions interrupt clinical workflow, while lower-confidence signals accumulate in non-interruptive feeds. Third, the research aligns distress markers with an **evolving digital CBT therapy**

landscape, where just-in-time adaptive interventions (JITAI) delivered through wearable devices are already demonstrating clinical efficacy in cardiac rehabilitation and depression management.

The Treatment Landscape Is Transforming

Pulsed Field Ablation: A Generational Advancement

The cardiac electrophysiology landscape is undergoing its most significant technological shift in a generation. Pulsed field ablation (PFA) - a minimally invasive procedure using targeted high-energy electrical pulses to treat irregular heart rhythms - has arrived as a transformative alternative to traditional thermal ablation. Unlike radiofrequency (heat) and cryoablation (cold), PFA selectively targets cardiac tissue while sparing surrounding structures, dramatically improving safety.

The pace of PFA innovation accelerated rapidly through 2025 and into 2026. Boston Scientific's FARAPULSE system was the first FDA-approved PFA platform for paroxysmal AF, with expanded labeling for persistent AF approved in early 2026. Abbott's Volt PFA System received FDA approval in December 2025, demonstrating 93.1% freedom from arrhythmia recurrence in paroxysmal AF and 81.9% in persistent AF at six months. Medtronic's PulseSelect, Johnson & Johnson's Varipulse, and Kardium's Globe catheter have all entered the market. This breadth of innovation means physicians have more tools than ever, patients have more options, and outcomes are improving across the board.

This is a profound advancement for patients. PFA are reducing recovery times and expanding access. Success rates are improving. Complication rates are declining. For the first time, there is real clinical momentum toward resolving the rhythm itself.

But Fixing the Rhythm Is Not Enough

Here is the critical insight that drives this project: **treating the arrhythmia does not automatically treat the psychological distress it caused.** Research documents 16.8% post-ablation depression rates. Patients who have lived with cardiac-related fear and avoidance do not automatically shed those patterns when the rhythm normalizes. ICD patients who have experienced shocks carry the trauma of those events regardless of subsequent device programming changes. The psychological imprint of an arrhythmia often outlasts the arrhythmia itself.

As PFA and other advanced ablation technologies succeed in treating more rhythm disorders more effectively, the population of patients who are cardiac-well but psychologically-unwell will grow. The ESC's 2025 consensus recognized exactly this: heart and mind care must become one. The investment in cardiac treatment technology must be matched by investment in understanding psychological outcomes. PFA and its successors fix the rhythm. Projects like MIRmade aim to contribute research toward understanding what happens after - and during, and before - the rhythm disrupts a life.

Why Now: The Convergence

1. Institutional mandate. The ESC 2025 Clinical Consensus Statement creates demand for psycho-cardio technology across 3,000+ European cardiology centers. The AHA's screening recommendations align. The institutional will exists; the technology does not.

2. Technology maturity. ECG foundation models (ECGFounder, pretrained on 10.77 million ECGs), clinical NLP (Bio_ClinicalBERT, trained on 880 million words of clinical notes), digital phenotyping platforms, and reinforcement learning for adaptive interventions have all matured independently. The tools arrived.

3. An emerging research frontier. While cardiac AI and digital mental health have each made tremendous strides independently, the application of predictive AI to the cardiac-psychological distress intersection remains in its earliest stages. MIRmade aims to contribute open-source research infrastructure that supports the broader psychocardiology community in exploring this frontier.

The Evidence for Integrated Care

Readmission reduction: Psychological distress is a documented independent predictor of cardiac readmission. CMS Hospital Readmissions Reduction Program penalties reach 3% of base Medicare DRG payments, creating a systemic incentive to identify and address the full spectrum of readmission risk factors - including psychological ones.

Healthcare consumption reduction: Särnholm et al. demonstrated a 56% reduction in healthcare consumption when cardiac-related fear and avoidance were treated with online CBT - sustained at 12 months. At EP clinic visit costs of $500-$2,000, the ROI is substantial.

Alignment with existing reimbursement pathways: The CMS CY2026 Physician Fee Schedule expanded Remote Therapeutic Monitoring (RTM) codes, creating reimbursement infrastructure for exactly the kind of continuous wearable-based monitoring and intervention delivery that psychocardiology research supports. These existing pathways mean that validated screening and monitoring tools can be sustainably integrated into clinical practice without requiring new billing categories.

GDP and workforce impact: Patients whose arrhythmia-related disability is driven by untreated psychological distress represent people who could return to fuller participation in work and life with appropriate support. Better understanding of distress markers can help identify where intervention would have the greatest impact - for patients and for the broader economy.

Arrhythmia Deserves More Attention

More than 100 million people worldwide live with cardiac rhythm disorders. One in three people will experience an arrhythmia in their lifetime. The rates of anxiety, depression, trauma, and suicidal ideation among them are not edge cases - they are the norm. And yet, the psychological dimension of their experience remains underserved.

The lack of visibility is why OneRhythm exists. While the treatment landscape is transforming with technology: **heart and mind care must become one.** Arrhythmia deserves more attention - more research, more collaboration, more compute applied to the problem. The healthcare system cost savings alone - reduced readmissions, reduced unnecessary utilization, reduced disability claims - along with positive GDP impact in reactivating parts of the labor force, make the case clear: this is a problem worth solving together.

This is no small problem.

OneRhythm will bring the visibility.
Project MIRmade aims to add to the compute.

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Nobody should fight an invisible bear alone.
MIRmade gives the bear a warning signal.

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